LIVE DEMO

Pantry Pal

Track your groceries, catch expiry dates, cut food waste.

01

Pantry

Add items and see what you have at home.

02

Calendar

See what expires and when, month by month.

03

Notifications

Get reminded before food goes bad.

04

Premium

Receipt scan, auto category, custom reminders, usage summary.

WHY PANTRY PAL

Problem and pain points

Food is forgotten, expires, and gets wasted. That costs money and adds to food waste.

FR-5

Forgetting what is at home

One list shows every active item.

FR-4

Food expires unnoticed

Reminders arrive before items expire.

FR-3

Hard to estimate expiry dates

Estimated from category and purchase date.

FR-6

Older food goes unused

Expiring-soon and expired items stand out.

FR-7, 8

Adding many items is slow

Premium receipt scan and auto categories.

FR-10, 11

Waste patterns stay hidden

Expired history and usage summary.

SRS · SECTION 3

Functional requirements

11 user stories. Premium features are marked.

- **FR-1** Account and Premium upgrade
- **FR-2** Manage grocery items
- **FR-3** Estimate expiration date
- **FR-4** Expiration reminders
- **FR-5** Display active items
- **FR-6** Show expiring and expired items
- **FR-7** Receipt or item scan (Premium)
- **FR-8** Auto category detection (Premium)
- **FR-9** Custom reminder settings (Premium)
- **FR-10** Expired history (7 days free)
- **FR-11** Usage summary (Premium)

SRS · SECTION 4

Non-functional requirements

NFR-1 · Performance
Save in 2 s; receipt scan in 5 s.

NFR-2 · Usability
First-time user adds or scans in 60 s.

NFR-3 · Reliability
Data persists after closing the app.

NFR-4 · Security / privacy
Users access only their own data.

NFR-5 · Availability
At least 99% uptime, excluding maintenance.

NFR-6 · Compatibility
Works on supported Android and iOS devices.

NFR-7 · Notification reliability
Alerts follow Free default or Premium setting.

NFR-8 · Maintainability
Modular: functions maintained independently.

NFR-9 · Data accuracy
Correct expiry estimates and category detection.

NFR-10 · Error handling
Clear message for invalid or missing input.

NFR-11 · Data accuracy (usage)
Each item counted once; counts reset monthly.

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Use cases

14 use cases cover all 11 FRs. The actor for all is the Household Member.

UC-1 Register	**FR-1**
UC-2 Manage account	**FR-1**
UC-3 Manage grocery items	**FR-2, 3**
UC-4 Scan item or receipt	**FR-7, 2, 8**
UC-5 Check scan access	**FR-7**
UC-6 Review scanned items	**FR-8, 2**
UC-7 Auto category detection	**FR-8**
UC-8 Receive expiration reminder	**FR-4**
UC-9 Customize reminders	**FR-9**
UC-10 View grocery inventory	**FR-5**
UC-11 View expiring or expired	**FR-6**
UC-12 View expired history	**FR-10**
UC-13 Upgrade to Premium	**FR-1**
UC-14 View usage summary	**FR-11**

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Golden Thread

Problem	Requirement	Built in	Test
Users need to manage their account	FR-1 Account	Registration, Login, Settings	TC-01
Forgetting what groceries they have	FR-2 Manage items	Pantry Dashboard, Add/Edit page	TC-02
Hard to estimate expiry dates	FR-3 Estimate expiry	Add/Edit Grocery page	TC-03
Forgetting upcoming expiry	FR-4 Reminders	Reminders page, Notifications	TC-04
Not knowing what is at home	FR-5 Active items	Pantry Dashboard, Active Items	TC-05
Missing expiring or expired food	FR-6 Expiry status	Expiring Soon, Expired sections	TC-06
Adding many items is slow	FR-7 Receipt scan	Scan Receipt, Premium Features	TC-07
Categorizing items takes time	FR-8 Auto category	Scan Review, Add/Edit pages	TC-08
Users need different lead times	FR-9 Reminder setup	Reminder Settings page	TC-09
Reviewing past expired food	FR-10 History	Grocery History page	TC-10
Understanding waste patterns	FR-11 Usage summary	Usage Summary page	TC-11

SRS · SECTION 8

AI usage log

6 entries, 22 Aug to 2 Nov 2026 · ChatGPT, Claude, Perplexity, Gemini, Readdy AI

Requirements and use cases

Asked for help modifying FRs, NFRs, and use cases. AI suggested them; we set the actor to Household Member and fixed FR and use-case links.

Checking our work

Asked AI to check the use-case diagram and requirements. It flagged mismatches and errors; we reviewed and fixed them.

Building the app

Asked AI to build the application. It listed features that did not match our FRs; we changed it to match.

Upgrading the app

Asked AI to upgrade the application. It returned the old version; we revised it to add more features.

We learned: FRs versus NFRs, linking FRs to use cases, and include/extend.